Coherent Quantum Feedback Control of a BEC Under Communication Constraints: Proposed Simulation and Analytical Study

Aman Chawla

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Abstract

BEC is a unique semiclassical/quantum system. We work with its simulations. The goal is to induce BEC formation using external controls under communication constraints. This proposal will test anytime information concepts in a quantum setting.

In this note the authors propose to control a BEC - induce solitons [1] in it - using coherent [2] or semiclassical control in the presence of a communication channel [3, 4, 5]. Our first step is to replicate the theorem on tracking an unstable stochastic process over a classical channel in the quantum setting, in order to understand the mechanics. The next step is to implement the BEC using a Gross-Pitaevskii Equation simulation. Then we will use the external potential term to control the BEC soliton evolution. The signals or commands to the external potential term will be generated via a closed-loop controller which will interface with the plant via the communication channel. In the final step both the plant and the channel will be fully quantized. This study as proposed, will help in testing the limits of anytime information theory in the quantum or semi-classical domain. Post-simulation an analytical study will be taken up to explore the fundamental limits.

References

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